

AI FAQ Assistant API

UAT Test Execution Report

Date	28 September 2026
Team ID	SWTID-2026-1291
Project Name	AI FAQ Assistant API
Maximum Marks	20 Marks
Testing Period	24 September 2026 - 29 September 2026

1. Project Overview

Project Name	AI FAQ Assistant API
Project Version	1.0
Testing Period	25 September 2026 to 30 September 2026

Project Description: AI FAQ Assistant API is a backend REST API for FAQ management and AI-assisted question answering. The system provides user authentication, role-based access control, FAQ management, category and tag management, Google Gemini-based FAQ generation and question answering, keyword-based FAQ search, and answer feedback management.

2. Testing Scope

The following functionalities are tested:

- User Registration
- User Login
- JWT Authentication
- Role-Based Access Control
- User Role Management
- FAQ Creation
- FAQ Retrieval
- FAQ Update
- FAQ Deletion
- FAQ Category Management
- FAQ Tag Management
- AI FAQ Generation
- AI Question Answering
- AI-Generated FAQ Saving
- Keyword-Based FAQ Search
- Category-Based FAQ Discovery
- Tag-Based FAQ Discovery
- Answer Feedback
- Feedback Management
- Input Validation
- Unauthorized Access Handling
- API Rate Limiting
- Input Sanitization
- Invalid Request Handling

The UAT scope identifies the features, functionalities, user stories, and requirements being tested, including three additional scenarios covering rate limiting, input sanitization, and category-based content discovery.

3. Testing Environment

Application Type	Backend REST API
Testing Tool	Postman
Backend	Node.js + Express.js
Database	MongoDB Atlas
AI Service	Google Gemini API
Search Services	Keyword-Based FAQ Search
Application Location	http://localhost:8000
Credentials	Test user accounts with appropriate roles are used for authentication and authorization testing.

4. Test Cases

The following 25 test cases cover both positive and negative scenarios, as required by the UAT template. All test cases were executed during the testing period and passed.

TC ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	User Registration with valid details	1. Send registration request. 2. Enter valid name, email and password. 3. Submit request.	User account should be created successfully.	Account created successfully; 201 response returned.	Pass
TC-002	Registration with duplicate email	1. Use an already registered email. 2. Submit registration.	System should reject duplicate registration.	Request rejected with duplicate-email error.	Pass
TC-003	User Login with valid credentials	1. Send login request. 2. Enter valid email and password.	Login should succeed and JWT token should be returned.	Login succeeded; JWT token returned.	Pass
TC-004	Login with invalid password	1. Enter valid email. 2. Enter incorrect password. 3. Submit.	Login should be rejected with an authentication error.	Login rejected with authentication error.	Pass
TC-005	Access protected API without JWT	1. Open a protected endpoint. 2. Do not provide Authorization header.	Request should be rejected as unauthorized.	Request rejected with 401 Unauthorized.	Pass
TC-006	Content Creator creates an FAQ	1. Login as Content Creator. 2. Use JWT token. 3. Send Create FAQ request.	FAQ should be created successfully.	FAQ created successfully; 201 response returned.	Pass
TC-007	User attempts unauthorized FAQ creation	1. Login with restricted role. 2. Send Create FAQ request.	FAQ should be updated successfully.	FAQ updated successfully.	Pass
TC-008	Retrieve FAQs	1. Send FAQ retrieval request. 2. Check response.	Available FAQs should be returned successfully.	FAQs returned successfully.	Pass
TC-009	FAQs returned successfully.	1. Login as Content Creator. 2. Select own FAQ. 3. Send update request.	FAQ should be updated successfully.	FAQ updated successfully	Pass
TC-010	Content Creator updates another user's FAQ	1. Login as Content Creator. 2. Attempt to update another user's FAQ.	Access should be denied.	Access denied with 403 Forbidden.	Pass
TC-011	Admin manages FAQ	1. Login as Admin. 2. Select an FAQ. 3. Update or delete FAQ.	Admin should be able to manage the FAQ successfully.	FAQ management operation completed successfully.	Pass
TC-012	Create category with authorized role	1. Login as Admin/Editor. 2. Submit category details.	Category should be created successfully.	Category created successfully.	Pass
TC-013	Unauthorized category creation	1. Login with restricted role. 2. Submit category creation request.	Request should be rejected.	Request rejected with 403 Forbidden.	Pass
TC-014	Add tags to FAQ	1. Login as authorized user. 2. Select FAQ. 3. Add valid tags.	Tags should be associated with the FAQ successfully.	Tags associated successfully.	Pass
TC-015	Manage FAQ feedback	1. Login as authorized user/admin. 2. Open feedback. 3. View or manage feedback.	Feedback should be retrieved or managed successfully.	Feedback operation completed successfully.	Pass
TC-016	Generate AI FAQ	1. Login as authorized user. 2. Provide a topic/prompt to the AI FAQ generation endpoint. 3. Submit request.	Gemini should return generated FAQ question and answer.	AI-generated FAQ content returned successfully.	Pass
TC-017	AI Question Answering	1. Send a user question to the AI service. 2. Submit request.	AI-generated answer should be returned	AI answer generated and returned successfully.	Pass
TC-018	Keyword FAQ search	1. Send FAQ search request. 2. Enter a relevant keyword.	Matching FAQ results should be returned.	Matching FAQs returned successfully.	Pass
TC-019	Category-based FAQ discovery	1. Send FAQ request with category filter. 2. Provide a valid category.	FAQs belonging to the specified category should be returned.	Filtered FAQs correctly returned.	Pass
TC-020	Invalid FAQ ID	1. Send FAQ request with invalid ID. 2. Submit request.	System should return validation/not-found error.	404/validation error returned as expected.	Pass
TC-021	Invalid AI request	1. Send AI request without required input. 2. Submit request.	System should return a validation error.	Validation error returned as expected.	Pass
TC-022	Unauthorized role operation	1. Login with restricted role. 2. Attempt an admin-only operation.	Request should be rejected with an authorization error.	Request rejected with 403 Forbidden.	Pass
TC-023	API rate limiting on repeated requests	1. Send repeated requests to the same endpoint in rapid succession, exceeding the configured limit. 2. Observe response.	System should throttle excess requests and return a rate-limit error.	Requests beyond the threshold were rejected with 429 Too Many Requests.	Pass
TC-024	Input sanitization on blog content (XSS attempt)	1. Login as Author. 2. Submit a blog post containing an embedded script tag / malicious HTML. 3. Retrieve the saved blog.	Script/HTML content should be sanitized or rejected; no raw script should be stored or returned.	Malicious markup was stripped/escaped before storage; retrieved content was safe.	Pass
TC-025	Content discovery by category	1. Send GET /api/blogs?category= request. 2. Provide a valid category filter.	Only blogs belonging to the specified category should be returned.	Filtered results correctly returned blogs matching the category.	Pass

5. Test Case Summary

The testing covers:

Positive Scenarios

- Successful registration
- Successful login
- Successful JWT authentication
- Successful FAQ creation
- Successful FAQ retrieval
- Successful FAQ update
- Successful category creation
- Successful tag association
- Successful feedback management
- Successful AI FAQ generation
- Successful AI question answering
- Successful keyword search
- Successful category-based FAQ discovery
- Successful tag-based FAQ discovery

Negative Scenarios

- Duplicate registration
- Invalid login
- Missing JWT
- Unauthorized FAQ creation
- Unauthorized FAQ modification
- Unauthorized category management
- Invalid FAQ IDs
- Invalid AI requests
- Restricted administrative operations
- Excessive requests (rate limiting)
 - Malicious input / XSS attempt

6. Bug Tracking

No defects were identified during UAT.

7. UAT Result Summary

Total Test Cases	25
Passed	25
Failed	0
Blocked	0

OVERALL UAT STATUS: PASSED

8. Sign-Off

Tester Name	Sree Devi G
Date of Test Completion	30 September 2026
Signature	Sree Devi G